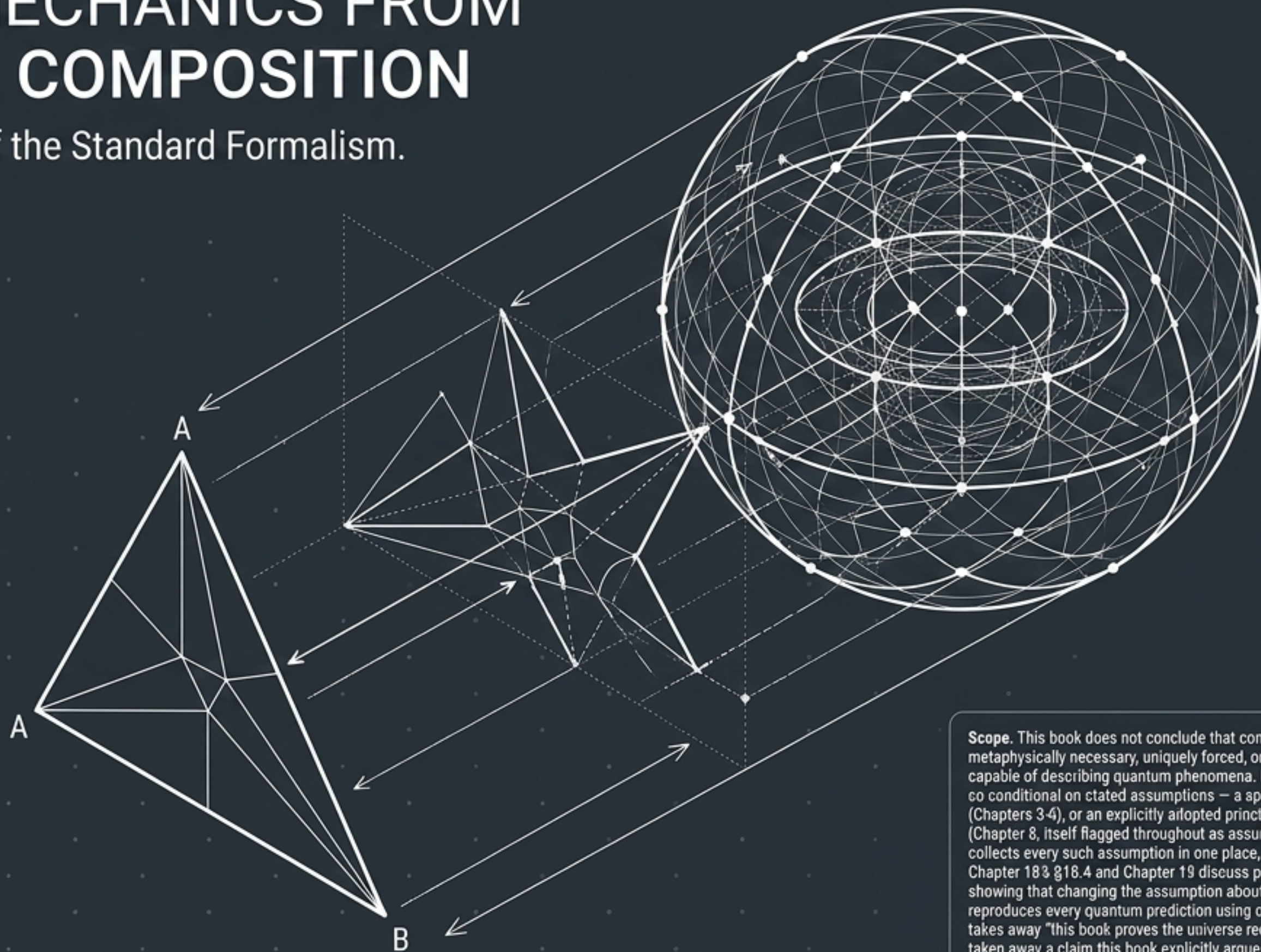


QUANTUM MECHANICS FROM ADMISSIBLE COMPOSITION

An Architectural Audit of the Standard Formalism.

We do not ask how to use standard quantum mechanics. We ask **exactly what it costs to build it.**

By enforcing strict constraints on probabilistic admissibility, we audit the mathematical foundations of physics to discover why nature insists on complex numbers, tensor products, and unitary dynamics.



SCOPE

Scope. This book does not conclude that complex numbers are metaphysically necessary, uniquely forced, or the only mathematics capable of describing quantum phenomena. Every result ruling out $\mathbb{R}$ does so conditional on stated assumptions – a specific admissibility constraint (Chapters 3-4), or an explicitly adopted principle such as local tomography (Chapter 8, itself flagged throughout as assumed, not derived). Chapter 18 collects every such assumption in one place, exactly four of them, and Chapter 18 §18.4 and Chapter 19 discuss published 2025-2026 work showing that changing the assumption about how systems compose reproduces every quantum prediction using only real numbers. A reader who takes away “this book proves the universe requires complex numbers” has taken away a claim this book explicitly argues against.

The Probability Baseline and Reversible Lifts

The Probability Simplex

Bare, nonnegative probabilities.
Constraint before content.

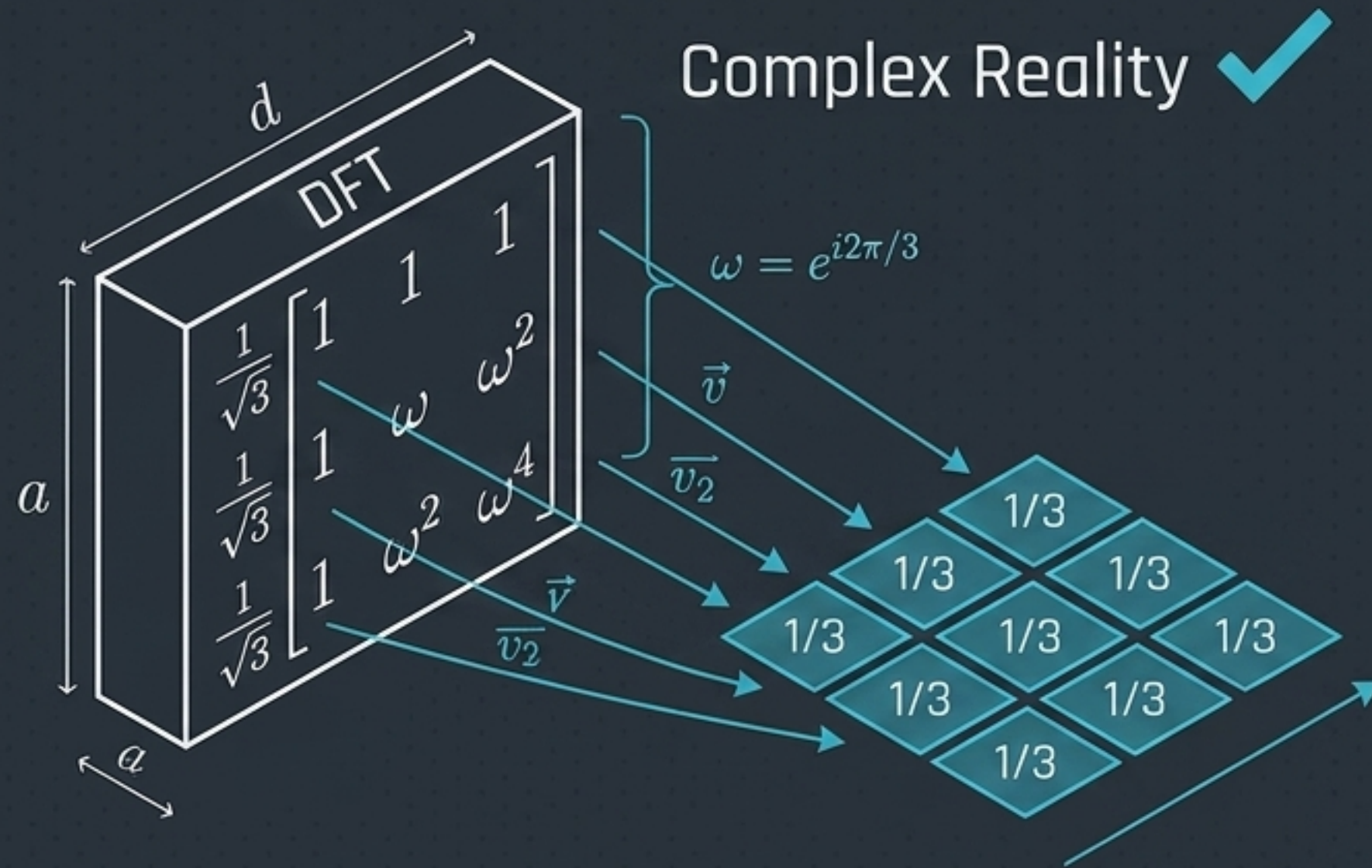
Bistochastic Evolution

Constraint: Reversibility imposed.
Transitions preserve Shannon entropy.

Unistochastic Lifts

Probability transitions perfectly
modeled by the squared moduli of
unitary amplitudes.

The n=3 Real Incompleteness Theorem

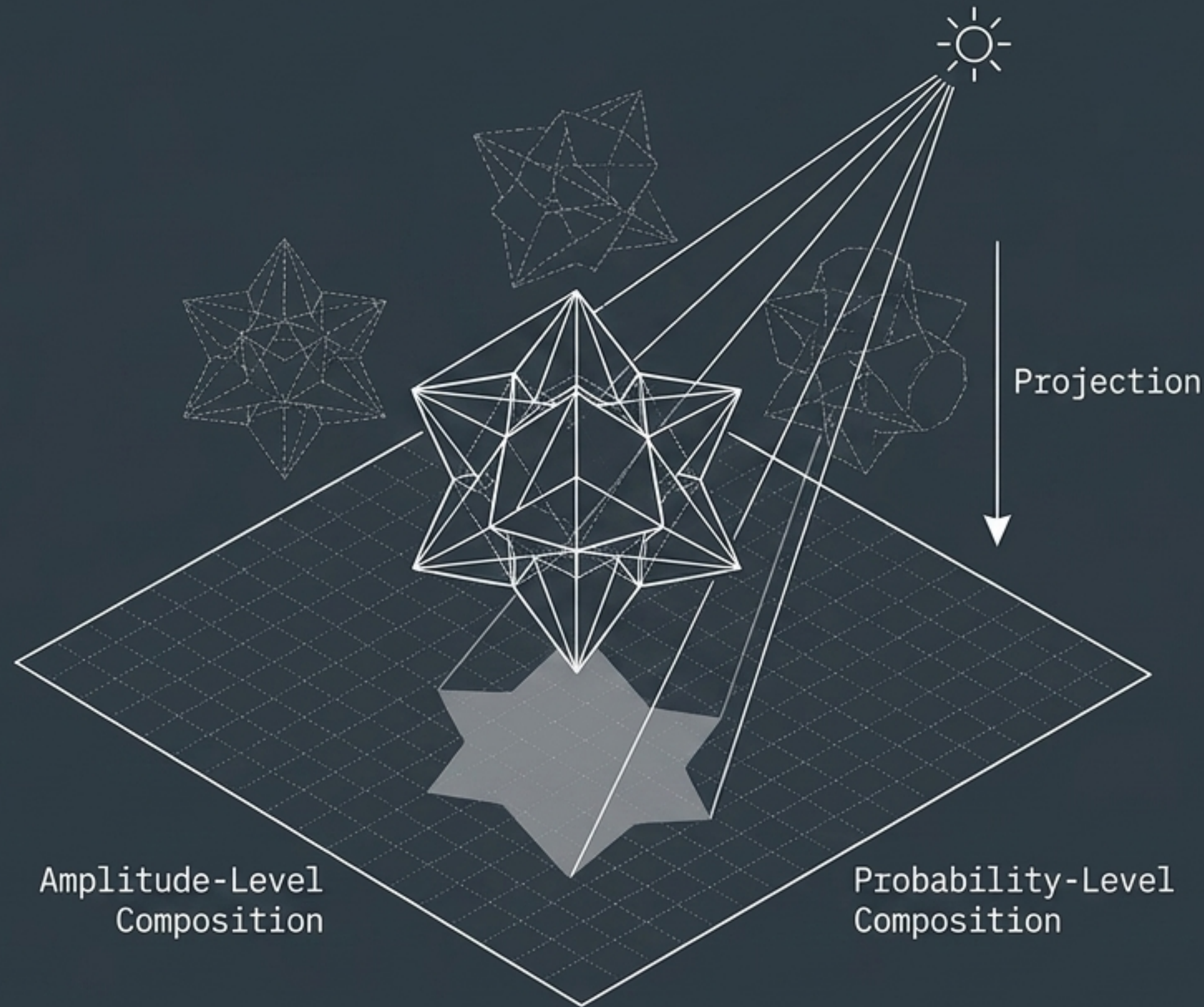


At $d=2$, Real and Complex unistochastic matrices degenerate. They are mathematically identical.

At $d=3$, the real framework structurally fails. No single real orthogonal matrix can reproduce the uniform probabilities generated by the complex DFT matrix. Complex numbers become a necessity, not a choice.

Scope. This is a fact about one mathematical object — a specific unistochastic matrix... not yet a claim about physical real-scalar field... nothing here has connected "this matrix" to "a physical measurement"...

Information Loss in the Probability Projection

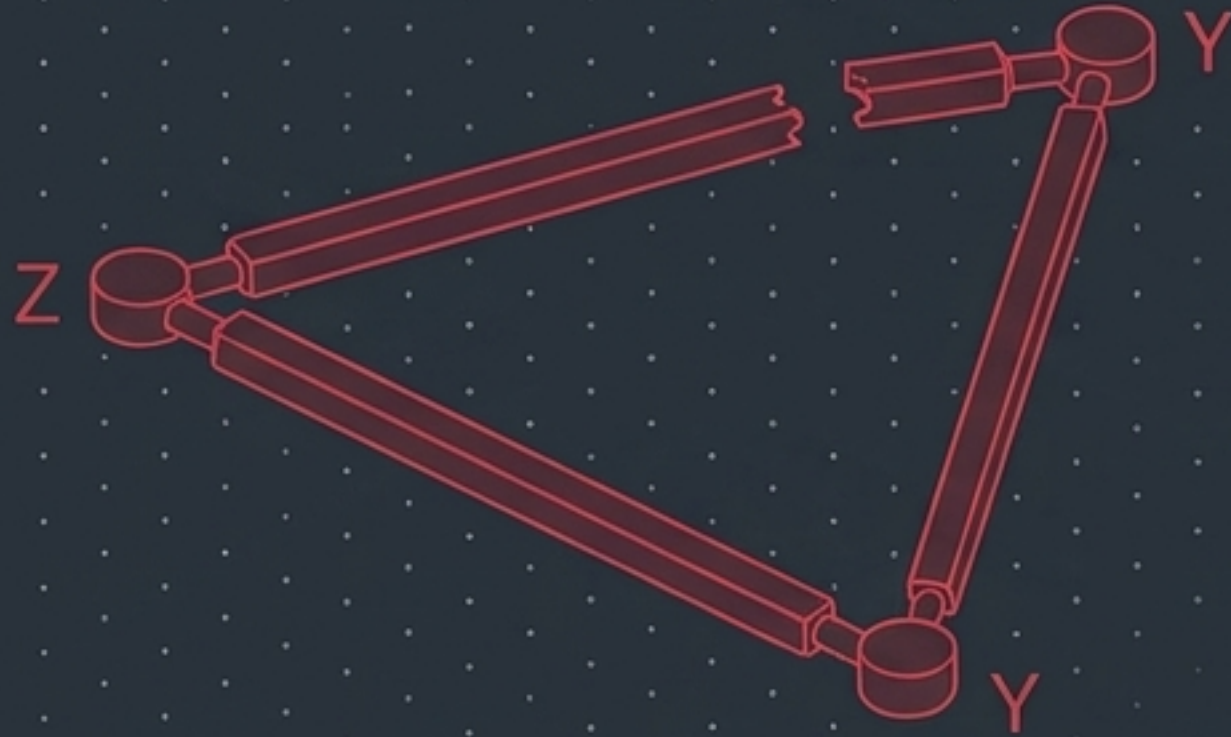


Information Loss in the Probability Projection

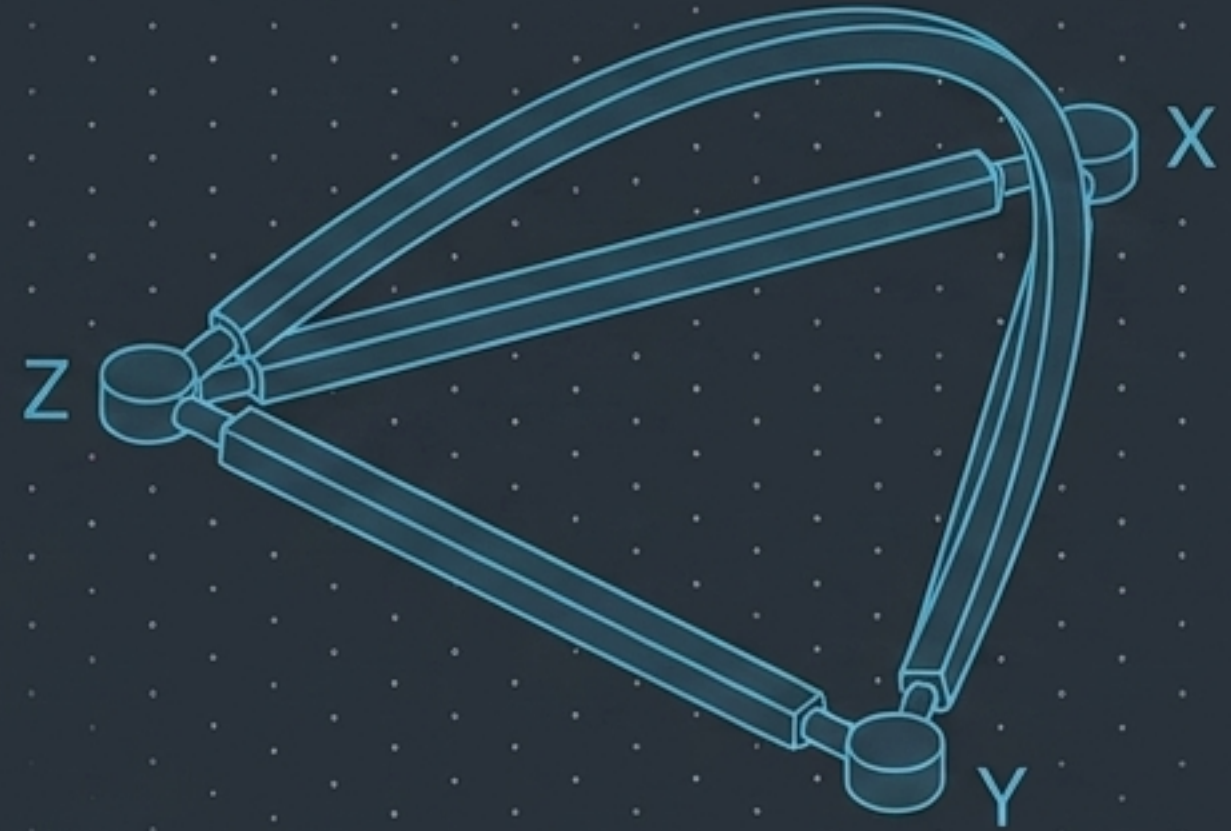
- The modulus-square map is not a homomorphism.
- Composing at the amplitude level and projecting generates interference cross-terms.
- Composing strictly at the probability level discards the geometry required to track this interference.

The Context Graph and the Cycle Obstruction

The Real Failure



The Complex Solution



- Edgewise Admissibility $\neq$ Global Realizability
- Individually admissible transitions may fail to close globally around a fundamental cycle.
- **The MUB Obstruction:** There are no three pairwise mutually unbiased orthonormal bases in Real space. Real numbers mathematically refuse to close the triangle.

Scope. This rules out Real space for one specific configuration — three mutually unbiased contexts — not for real-valued physics in general. Treat "real numbers cannot do this" as scoped to this exact triangle, this exact dimension, this exact data.

The Dimensional Gearbox: Local Tomography



How must independently accessible systems compose? If a joint state is exactly as informative as a tomographically complete set of local measurements, the tensor product mathematically forces Complex numbers as the unique “Goldilocks” fit.

Scope. “Unique” above means unique among $\mathbb{R}$, $\mathbb{C}$, $\mathbb{M}$, under this specific dimension-counting test, using the ordinary Kronecker product. It is not a claim that $\mathbb{C}$ is the only conceivable scalar structure for physics, and it is not unconditional: local tomography itself is an assumed principle this chapter tests rather than derives (this chapter’s closing ledger says so explicitly), and Chapter 18’s §18.4 discusses published work in which changing the composition rule (not the scalar field) lets a real theory pass an equivalent test. Read this theorem as “given local tomography and the Kronecker product, $\mathbb{C}$ is the unique fixed point” — every word of that qualification is load-bearing.

The Scalar Field Stress Test

Field	Single System ($d \geq 3$)	Composite Dimension Mismatch	MUB Subspace Behavior
Real	Fails Orthostochastic match	Misses parameters (>0)	Restricted ($d/2 + 1$)
Complex	Exact closure	Perfect fit (0)	Exact ($d + 1$)
Quaternionic	Closes	Overcounts (<0)	Over-allows ($2d + 1$)

This audit tests Real, Complex, and Quaternionic fields against escalating structural requirements. Complex quantum mechanics is not an arbitrary choice; it is the unique survivor of this architectural inspection.

Identifying the Invisible Real Dimension

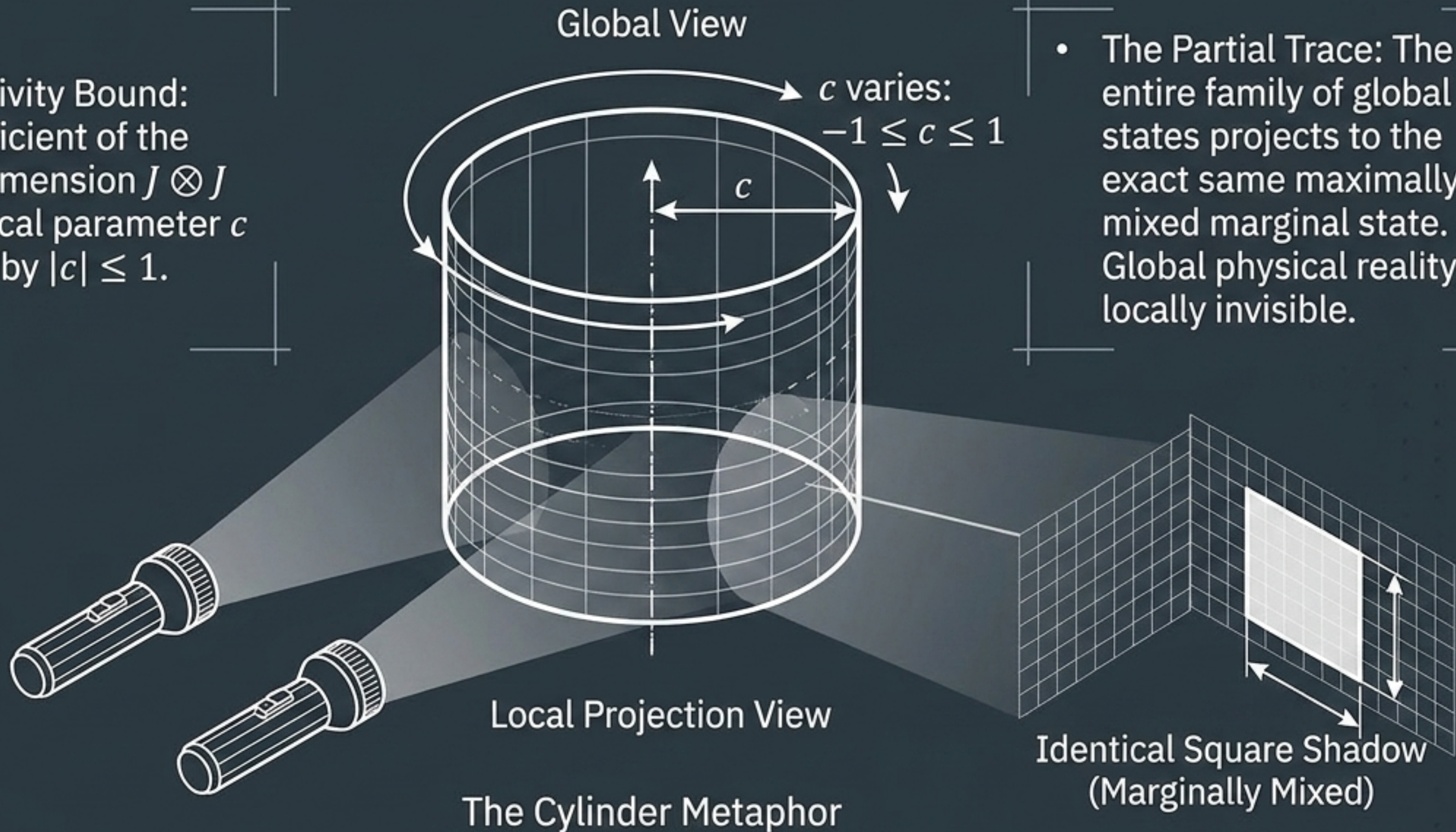


For two real qubits, local product bases provide 9 dimensions. The composite space requires 10. The missing tenth dimension is spanned by $J \otimes J$.

Real composite systems carry a genuinely non-local degree of freedom built from two individually-inadmissible local generators whose product happens to be perfectly admissible.

Positivity Bounds and Marginal Invisibility

- The Positivity Bound: The coefficient of the hidden dimension $J \otimes J$ is a physical parameter c bounded by $|c| \leq 1$.

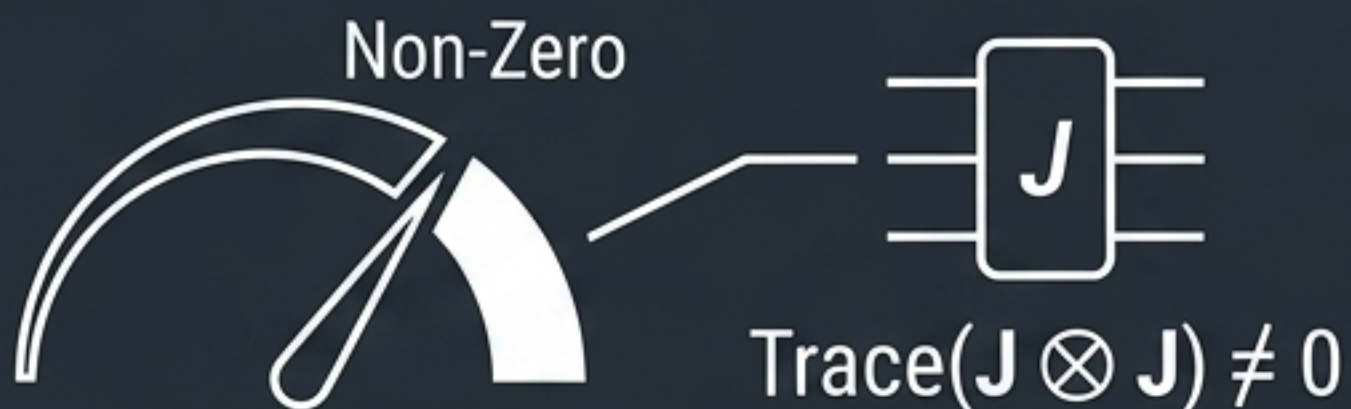


- The Partial Trace: The entire family of global states projects to the exact same maximally mixed marginal state. Global physical reality is locally invisible.

The Relativity of Entanglement

Viewed through Real Lens

ENTANGLED



Viewed through Complex Lens

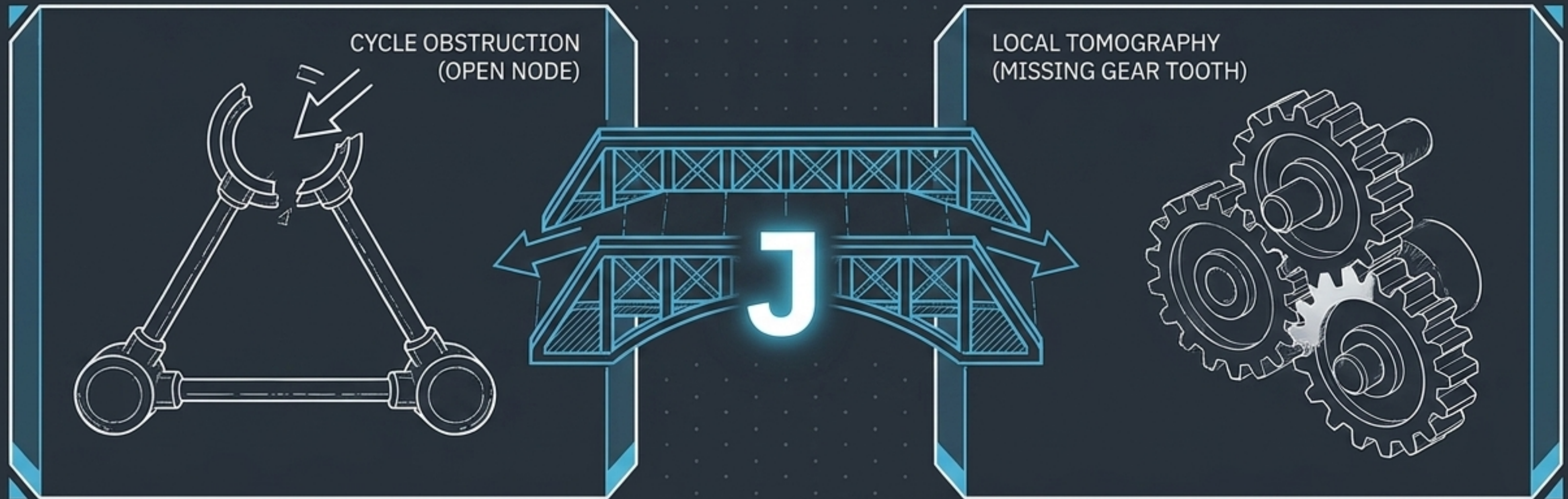
SEPARABLE

$$\rho = \begin{cases} p_1 |\psi_1\rangle\langle\psi_1| + p_2 |\psi_2\rangle\langle\psi_2| + \\ p_3 |\psi_3\rangle\langle\psi_3| + p_4 |\psi_4\rangle\langle\psi_4| \end{cases}$$

Separability is not intrinsic to a state matrix. It is relative to the scalar field allowed to compose the local subsystems. The identical matrix is Real-entangled but Complex-separable.

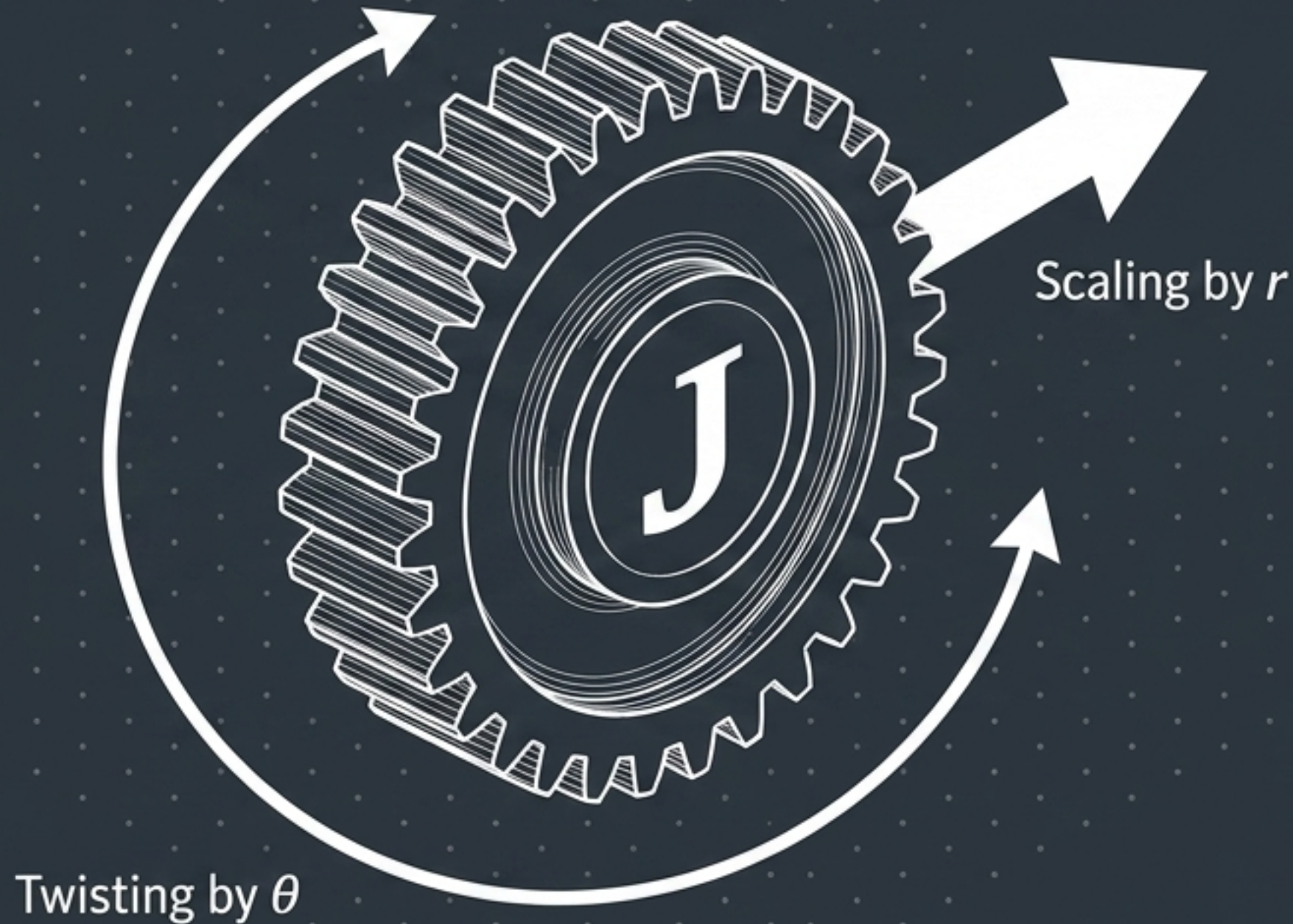
Scope. This is a precise claim about one family of matrices under Definition 12.1, not a general verdict that “entanglement isn’t physically real.” It does not say every entangled state in ordinary complex quantum mechanics becomes separable under some other field, and it does not say separability judgments are arbitrary or a matter of choice once a scalar field is fixed — within ordinary complex quantum mechanics, $\rho(c)$ ’s entanglement status (for whatever states arise there) is perfectly well-defined. What is field-relative is the comparison across candidate theories with different admissible local product states, which is the question this book’s Parts I–II were built to ask.

Synthesis: Two Sides of the Same Coin



The degree of freedom that makes complex numbers structurally necessary for local tomography ($\mathbf{J} \otimes \mathbf{J}$) is the exact same rotational degree of freedom that hides in the margins of entanglement and allows cyclic measurement contexts to smoothly glue together globally.

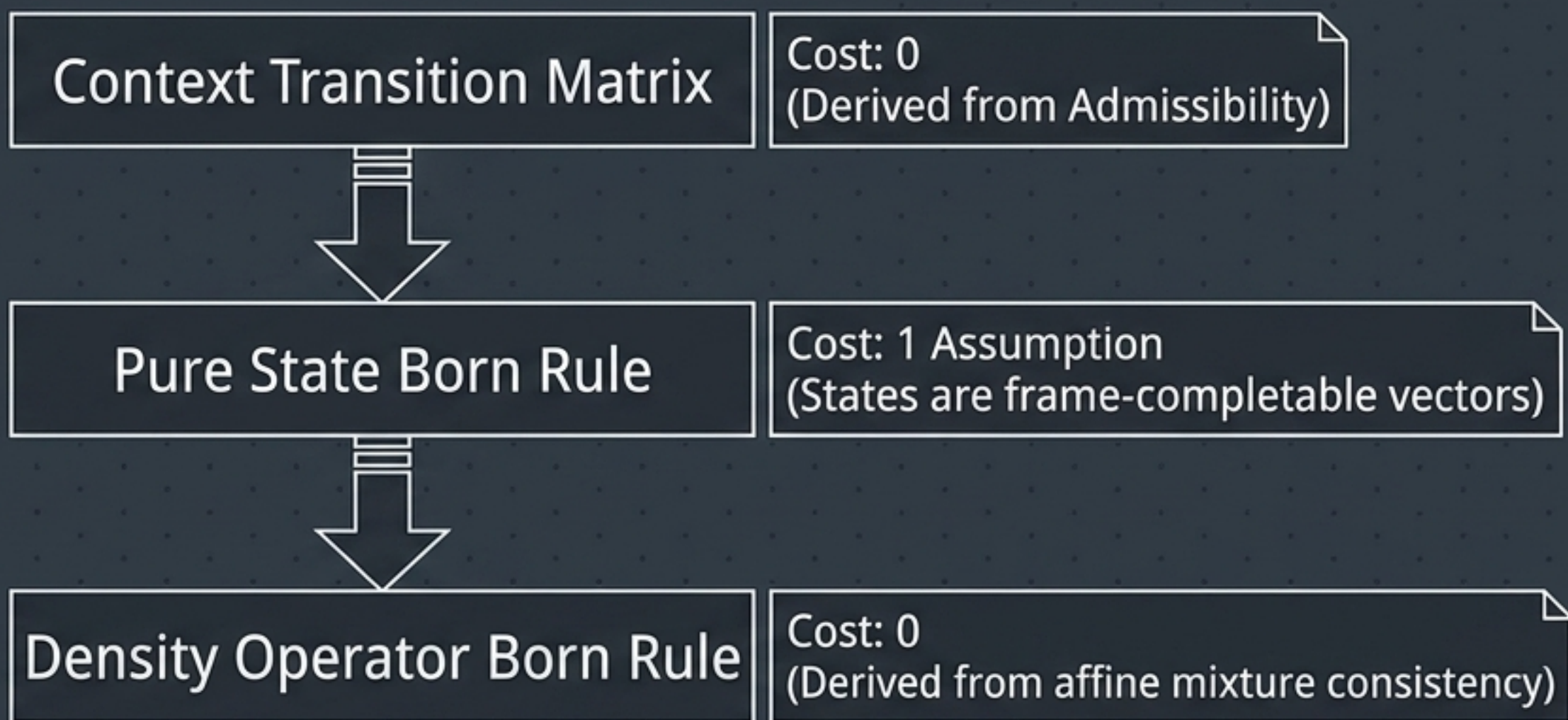
Amplitwists: The Physicality of i



Complex numbers are not metaphysical necessities or mathematical ghosts. The unit i is exactly physically equivalent to the real 2×2 antisymmetric matrix J . Multiplication by z is merely an amplitwist—simultaneous amplitude scaling and angular rotation. It is the fundamental physical gear required to process coherent transitions.

Scope. “The right invariant” is a claim about J , not a claim that complex scalars are the only way to write it. The whole point of this chapter is that J is expressible equally well as a literal 2×2 real matrix — the convergence above is evidence that this book’s earlier chapters were tracking a real structural fact, not evidence that the fact must be written using complex numbers specifically.

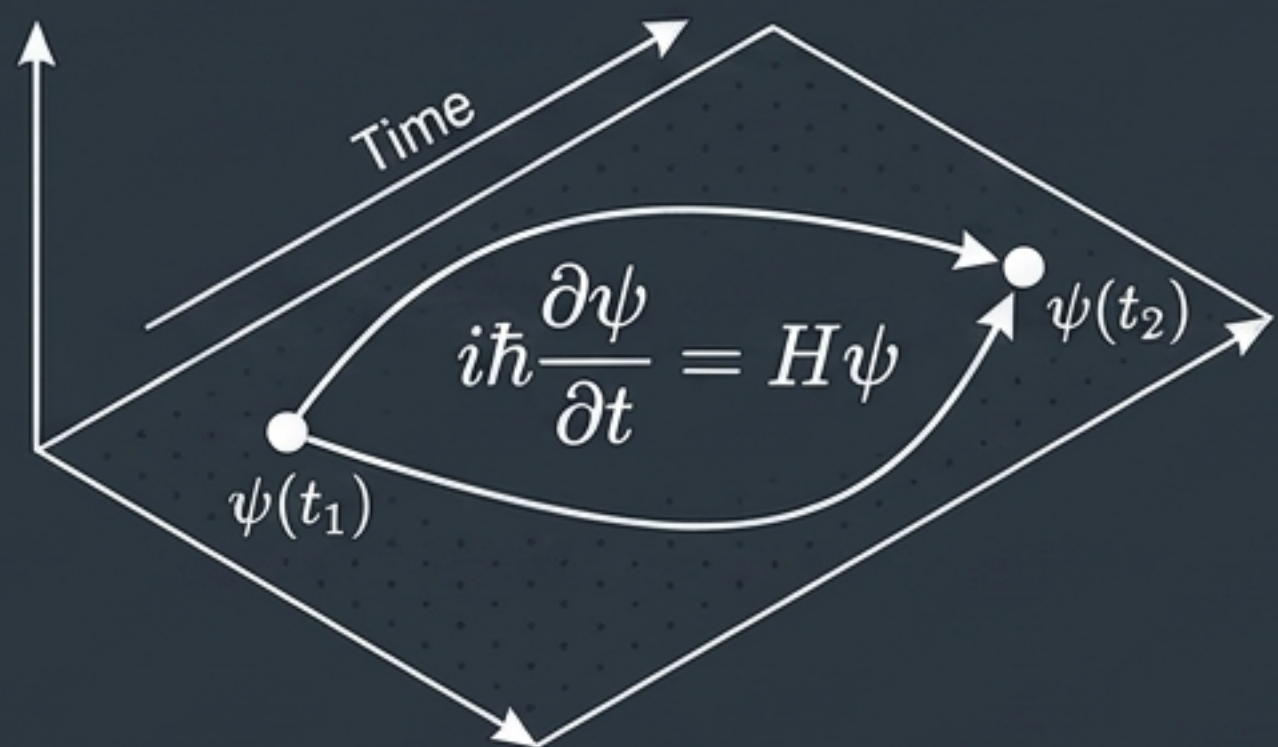
The Born Hierarchy Cascade



The Born rule is not a monolithic postulate. It is a hierarchy of derivations. Once a state is operationally identified with a frame coordinate, textbook probabilities cascade inevitably from the context-transition identity.

Recovering Dynamics and Generalized Measurement

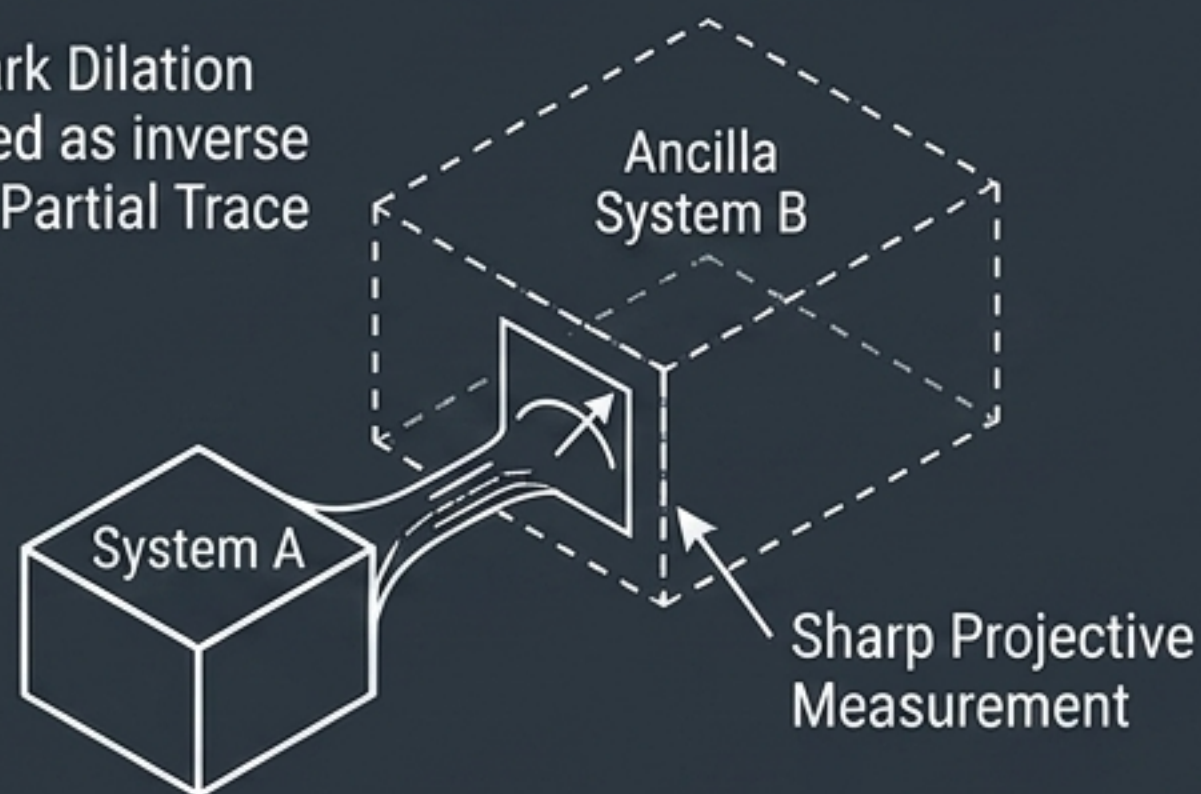
Dynamics



Schrödinger's Equation: The inevitable infinitesimal form of imposing continuous, reversible time evolution on the unitary group.

POVMs

Naimark Dilation mapped as inverse to the Partial Trace



POVMs: Generalized measurements are simply sharp projective measurements performed on a larger, dilated system-plus-ancilla architecture.

The Final Assumption Ledger

Derived from Constraints	The 4 Postulates Cost
- The Probability Simplex & Entropy behavior - Complex numbers as structural necessities - The Born Rules & Tensor Products - Entanglement & Partial Traces - Schrödinger's Equation & POVMs	1. Local tomography 2. Pure states are frame-completable vectors 3. Measurement outcomes are real-valued 4. Time evolution is a continuous unitary group

Standard quantum mechanics is not an arbitrary list of axioms. An 80-page mathematical audit reduces the entire standard textbook formalism to exactly four operational physical assumptions layered on top of bare probabilistic admissibility.